EXPERIMENT 9

Demonstration of Temporal Mining Techniques.

AIM

To demonstrate temporal data mining techniques including time series analysis, trend detection, seasonality decomposition, and sequential pattern mining.

PROCEDURE

1. Import pandas, numpy, matplotlib, and statsmodels libraries.
2. Load a time-stamped dataset (e.g., COVID-19 daily data or air passenger data).
3. Parse and set the date column as the index.
4. Resample data to monthly frequency and compute aggregates.
5. Plot the time series to observe overall trends.
6. Apply rolling window smoothing (7-day or 30-day moving average).
7. Perform seasonal decomposition using statsmodels seasonal_decompose().
8. Plot trend, seasonality, and residual components.
9. Identify peak periods and significant temporal patterns.
10. Apply sequence pattern mining to extract temporal rules.
11. Summarize findings of temporal patterns.

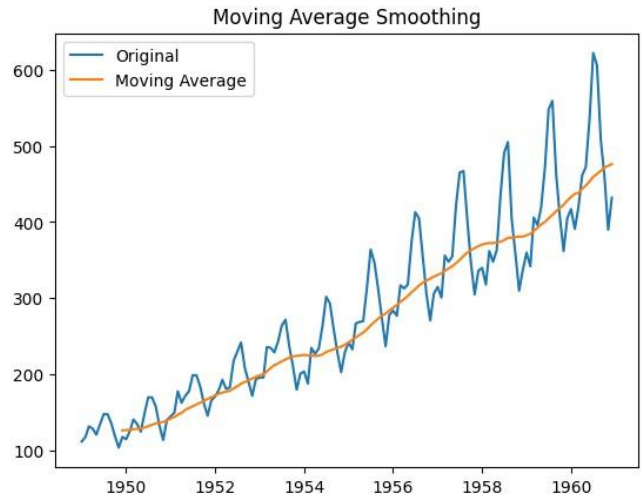
CODE

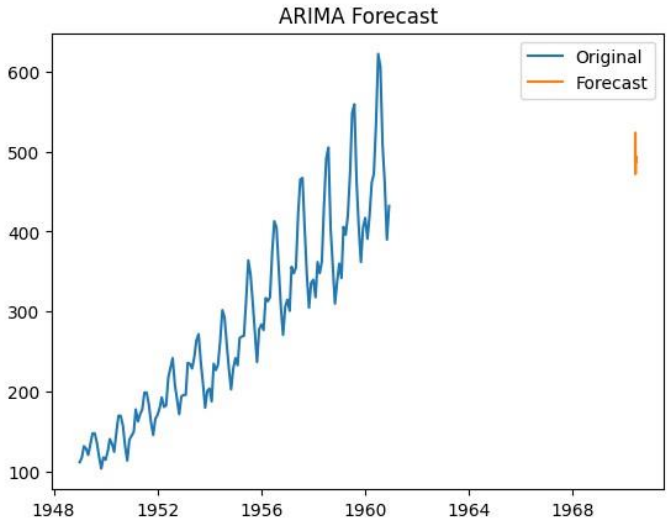
```
import pandas as pd import numpy as np import
matplotlib.pyplot as plt from statsmodels.tsa.seasonal
import seasonal_decompose

# Load temporal dataset
data = pd.read_csv('worldometer_coronavirus_daily_data.csv')
data['date'] = pd.to_datetime(data['date']) data =
data.dropna()

# Filter for a single country country_data =
data[data['country'] == 'USA'].copy()
country_data.set_index('date', inplace=True)
country_data = country_data.sort_index()

# Monthly Resampling monthly =
country_data['daily_new_cases'].resample('M').sum()

# Plot Time Series plt.figure(figsize=(12,
5))
plt.plot(monthly, label='Monthly New Cases')
plt.title('COVID-19 Monthly New Cases - USA')
plt.xlabel('Date') plt.ylabel('Cases')
plt.legend()
plt.show()

# Rolling Average rolling_avg =
country_data['daily_new_cases'].rolling(window=30).mean()
plt.figure(figsize=(12, 5)) plt.plot(country_data['daily_new_cases'],
```

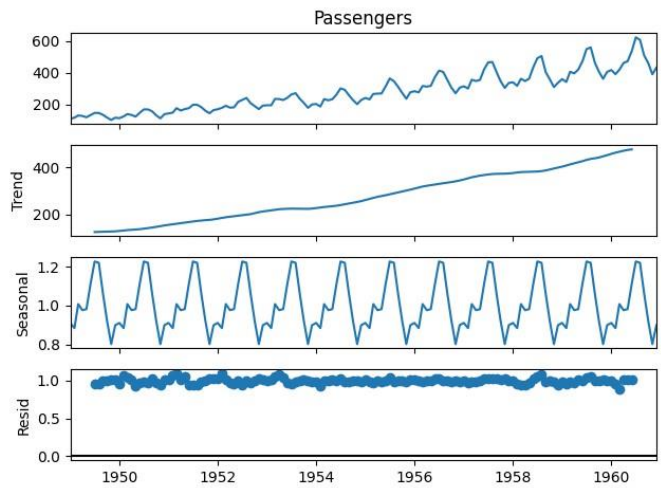
```
alpha=0.4, label='Daily Cases') plt.plot(rolling_avg, label='30-Day
Rolling Average', color='red') plt.title('Rolling Average - Daily
New Cases') plt.legend()
plt.show()

# Seasonal Decomposition decomposition = seasonal_decompose(monthly,
model='additive', period=12) decomposition.plot()
plt.suptitle('Seasonal Decomposition of Monthly COVID Cases')
plt.tight_layout() plt.show()
```

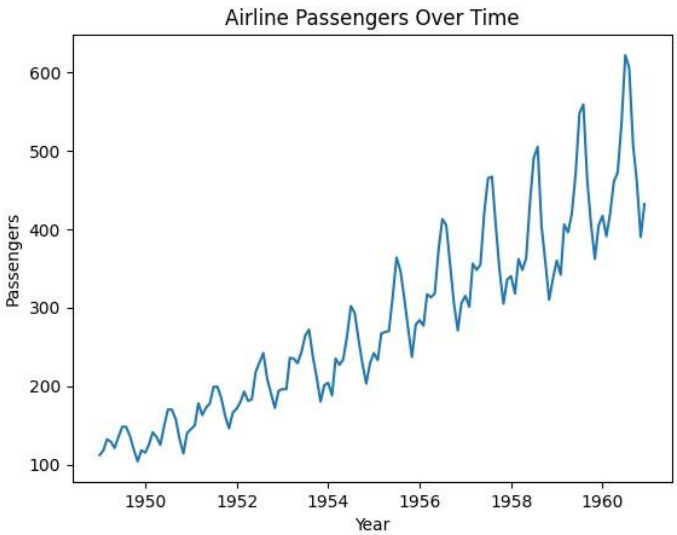
OUTPUT

```
Monthly resampled time series plotted for USA COVID-19 cases.
30-day rolling average smoothed the daily fluctuations.
Seasonal Decomposition revealed:
- Trend component: Rising peak in 2020-2021, decline afterwards.
- Seasonal component: Periodic wave patterns visible.  - Residual: Small
  irregular noise.
```









RESULT

Temporal mining techniques successfully revealed time-based patterns in the dataset. Rolling averages smoothed noise for clearer trend visibility. Seasonal decomposition separated the series into trend, seasonality, and residual components, enabling actionable insights about temporal behavior.